

D.2-4

Let A and B be $n \times n$ matrices such that $AB = I$. Prove that if A' is obtained from A by adding row j into row i , where $i \neq j$, then subtracting column i from column j of B yields the inverse B' of A' .

D.2-5

Let A be a nonsingular $n \times n$ matrix with complex entries. Show that every entry of A^{-1} is real if and only if every entry of A is real.

D.2-6

Show that if A is a nonsingular, symmetric, $n \times n$ matrix, then A^{-1} is symmetric. Show that if B is an arbitrary $m \times n$ matrix, then the $m \times m$ matrix given by the product BAB^T is symmetric.

D.2-7

Prove Theorem D.2. That is, show that a matrix A has full column rank if and only if $Ax = 0$ implies $x = 0$. (*Hint*: Express the linear dependence of one column on the others as a matrix-vector equation.)

D.2-8

Prove that for any two compatible matrices A and B ,

$$\text{rank}(AB) \leq \min \{ \text{rank}(A), \text{rank}(B) \},$$

where equality holds if either A or B is a nonsingular square matrix. (*Hint*: Use the alternate definition of the rank of a matrix.)

Problems**D-1 Vandermonde matrix**

Given numbers $x_0, x_1, \dots, x_{n-1}$, prove that the determinant of the *Vandermonde matrix*

$$V(x_0, x_1, \dots, x_{n-1}) = \begin{pmatrix} 1 & x_0 & x_0^2 & \dots & x_0^{n-1} \\ 1 & x_1 & x_1^2 & \dots & x_1^{n-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n-1} & x_{n-1}^2 & \dots & x_{n-1}^{n-1} \end{pmatrix}$$

is

$$\det(V(x_0, x_1, \dots, x_{n-1})) = \prod_{0 \leq j < k \leq n-1} (x_k - x_j).$$

(Hint: Multiply column i by $-x_0$ and add it to column $i + 1$ for $i = n - 1, n - 2, \dots, 1$, and then use induction.)

D-2 Permutations defined by matrix-vector multiplication over $GF(2)$

One class of permutations of the integers in the set $S_n = \{0, 1, 2, \dots, 2^n - 1\}$ is defined by matrix multiplication over $GF(2)$, the Galois field of two elements. For each integer $x \in S_n$, we view its binary representation as an n -bit vector

$$\begin{pmatrix} x_0 \\ x_1 \\ x_2 \\ \vdots \\ x_{n-1} \end{pmatrix},$$

where $x = \sum_{i=0}^{n-1} x_i 2^i$. If A is an $n \times n$ matrix in which each entry is either 0 or 1, then we can define a permutation mapping each value $x \in S_n$ to the number whose binary representation is the matrix-vector product Ax . All this arithmetic is performed over **$GF(2)$** : all values are either 0 or 1, and with one exception, the usual rules of addition and multiplication apply. The exception is that $1 + 1 = 0$. You can think of arithmetic over $GF(2)$ as being just like regular integer arithmetic, except that you use only the least-significant bit.

As an example, for $S_2 = \{0, 1, 2, 3\}$, the matrix

$$A = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$$

defines the following permutation π_A : $\pi_A(0) = 0$, $\pi_A(1) = 3$, $\pi_A(2) = 2$, $\pi_A(3) = 1$. To see why $\pi_A(3) = 1$, observe that, working in $GF(2)$,

$$\begin{aligned}\pi_A(3) &= \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \\ &= \begin{pmatrix} 1 \cdot 1 + 0 \cdot 1 \\ 1 \cdot 1 + 1 \cdot 1 \end{pmatrix} \\ &= \begin{pmatrix} 1 \\ 0 \end{pmatrix},\end{aligned}$$

which is the binary representation of 1.

For the remainder of this problem, we'll work over $GF(2)$, and all matrix and vector entries will be 0 or 1. Define the **rank** of a 0-1 matrix (a matrix for which each entry is either 0 or 1) over $GF(2)$ the same as for a regular matrix, but with all arithmetic that determines linear independence performed over $GF(2)$. We define the **range** of an $n \times n$ 0-1 matrix A by

$$R(A) = \{y : y = Ax \text{ for some } x \in S_n\},$$

so that $R(A)$ is the set of numbers in S_n that are produced by multiplying each value $x \in S_n$ by A .

- a.** If r is the rank of matrix A , prove that $|R(A)| = 2^r$. Conclude that A defines a permutation on S_n only if A has full rank.

For a given $n \times n$ matrix A and a given value $y \in R(A)$, we define the **preimage** of y by

$$P(A, y) = \{x : Ax = y\},$$

so that $P(A, y)$ is the set of values in S_n that map to y when multiplied by A .

- b.** If r is the rank of $n \times n$ matrix A and $y \in R(A)$, prove that $|P(A, y)| = 2^{n-r}$.

Let $0 \leq m \leq n$, and suppose that we partition the set S_n into blocks of consecutive numbers, where the i th block consists of the 2^m numbers

$i2^m, i2^m + 1, i2^m + 2, \dots, (i+1)2^m - 1$. For any subset $S \subseteq S_n$, define $B(S, m)$ to be the set of size- 2^m blocks of S_n containing some element of S . As an example, when $n = 3$, $m = 1$, and $S = \{1, 4, 5\}$, then $B(S, m)$ consists of blocks 0 (since 1 is in the 0th block) and 2 (since both 4 and 5 belong to block 2).

c. Let r be the rank of the lower left $(n - m) \times m$ submatrix of A , that is, the matrix formed by taking the intersection of the bottom $n - m$ rows and the leftmost m columns of A . Let S be any size- 2^m block of S_n , and let $S' = \{y : y = Ax \text{ for some } x \in S\}$. Prove that $|B(S', m)| = 2^r$ and that for each block in $B(S', m)$, exactly 2^{m-r} numbers in S map to that block.

Because multiplying the zero vector by any matrix yields a zero vector, the set of permutations of S_n defined by multiplying by $n \times n$ 0-1 matrices with full rank over $GF(2)$ cannot include all permutations of S_n . Let's extend the class of permutations defined by matrix-vector multiplication to include an additive term, so that $x \in S_n$ maps to $Ax + c$, where c is an n -bit vector and addition is performed over $GF(2)$. For example, when

$$A = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$$

and

$$c = \begin{pmatrix} 0 \\ 1 \end{pmatrix},$$

we get the following permutation $\pi_{A,c}$: $\pi_{A,c}(0) = 2$, $\pi_{A,c}(1) = 1$, $\pi_{A,c}(2) = 0$, $\pi_{A,c}(3) = 3$. We call any permutation that maps $x \in S_n$ to $Ax + c$, for some $n \times n$ 0-1 matrix A with full rank and some n -bit vector c , a **linear permutation**.

d. Use a counting argument to show that the number of linear permutations of S_n is much less than the number of permutations of

S_n .

- e.* Give an example of a value of n and a permutation of S_n that cannot be achieved by any linear permutation. (*Hint:* For a given permutation, think about how multiplying a matrix by a unit vector relates to the columns of the matrix.)

Appendix notes

Linear-algebra textbooks provide plenty of background information on matrices. The books by Strang [422, 423] are particularly good.